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A GPU/CPU co-designed engine lowers the compute cost of repository-scale proteomics reanalysis: development-set evaluation under a pre-registered protocol

Abstract

Public repositories hold tens of thousands of proteomics projects, but uniform reanalysis is limited by compute cost and by error control that degrades when results are pooled. Here, we describe Vulcan, an engine that keeps a proteome index resident on one GPU while CPU cores decode vendor files, evaluated under a pre-registered protocol. On development datasets, some used for tuning, Vulcan processed RAW files to protein groups at 4,570 runs per hour and 9.9 CPU core-seconds per run (GPU time excluded), about 7–12-fold fewer than a Sage search from mzML alone (Sage at 4–32 threads, CPU only). Peptidoform identifications were about 1% below (0.990-fold) the strongest comparator at matched entrapment-estimated error. Across up to 1,000 public runs, global competition kept estimated peptide error near 1%, whereas naively merged per-run lists reached 9.3%. However, the frozen scorer failed pre-registered null-spectrum tests, and the sealed held-out evaluation was interrupted by a node failure, so generalization and error control remain unvalidated.

Introduction

Public proteomics repositories have grown into a large and mostly unused resource. PRIDE alone now archives tens of thousands of projects[1], and joint reanalysis of such data has produced reference sets of peptides and post-translational modification (PTM) sites that no single study could generate[2–4]. Reanalysis is presented in these studies as a means to a reference rather than as the goal, but it is the step that limits how often such references can be rebuilt. Each run must be converted from a vendor format, searched against a proteome, rescored, filtered for errors and quantified, and current pipelines spend hundreds of CPU core-seconds per run on these steps[2,5].

Search engines have become fast. Fragment-index engines such as MSFragger[6] and Sage[5] search a run in seconds on a multi-core CPU, and chimeric search can add 15–57% peptides by explaining co-isolated precursors[7,8]. Graphics processing units (GPUs) have been tried for database search, but reported end-to-end gains have been modest, and in one open framework the GPU scoring step was slower than the multithreaded CPU path because it was memory-bound[9]. At repository scale the search itself is no longer the main cost. File conversion from RAW to mzML, per-run rescoring with predicted spectra[10,11] and protein inference together dominate, and the per-run overhead of starting a process and loading an index becomes significant when runs are small.

Error control is the second limit. Target–decoy competition is the standard way to estimate the false discovery rate (FDR) of peptide identifications[12], but it becomes harder to apply as data are pooled. When per-dataset 1% lists were merged into MassIVE-KB, the library precursor FDR reached 28%[4]. Classic protein-level FDR over-counts decoys as data grow[13], and picked protein-group FDR was needed to keep ProteomicsDB-scale inference calibrated[14]. Learned rescorers such as Percolator and mokapot add identifications[15,16] but can fail to control FDR in some settings, notably when runs are combined[17]. Entrapment experiments, in which foreign or shuffled sequences are searched alongside the real proteome, have recently shown that several widely used tools, mostly for data-independent acquisition, exceed their nominal FDR in some settings[18]. For repository-scale reanalysis, error control therefore needs to be tested, not assumed, and the tests need to be fixed before results are seen.

Here, we describe Vulcan, a data-dependent acquisition (DDA) proteomics engine in which each processing stage runs on the device that finishes the whole pipeline soonest. The proteome index stays resident on one A100 GPU in a long-lived process, while 32 CPU cores decode RAW files directly and run statistics. We evaluated the engine under a statistical protocol that was frozen before any held-out data were accessed. It specified entrapment arms, null-spectrum gates for learned scorers and a single sealed look at held-out test sets. All results reported here come from development datasets, several of which were used for method selection or tuning. The frozen default scorer failed three of the pre-registered null-spectrum gates, later attempts to re-test it were not evaluable or were not run, and the single sealed look was interrupted when the compute node was wiped. Its outputs were deleted unread. We therefore present the work as an engineering result with a stated evaluation protocol and a record of its negative outcomes, not as a validated method.

Results

Placing each stage on the device that finishes the pipeline first

Vulcan is organized as seven layers that operate on columnar tables (Fig. 1a). RAW files are decoded in-process with the vendor reader into an Arrow store that keeps all MS1 and MS2 centroids (L0). A per-run pilot pass estimates mass calibration and the activation and analyser class (L0.5). The proteome index (L1) is content-hashed and stores compact peptidoform records with paired decoys and entrapment sequences. It is laid out as a tiled inverted fragment index on the GPU. Candidate generation (L2) and exact rescoring with label-blind and predicted-spectrum features (L3) run as CUDA kernels. Statistics (L4), cross-run quantification (L5) and repository-scale FDR (L6) run on the CPU.

Two design rules shape the system. First, the search and scoring layers never see target, decoy or entrapment labels. A test (G0) swaps the labels of every target–decoy pair and permutes storage order, and requires every L2 and L3 output to be bit-identical. G0 passes for closed, chimeric, ion-trap collision-induced dissociation (IT-CID), cross-linking and semi-specific search; for the chimeric pipeline it was confirmed at the final freeze commit. Second, work is placed by measured cost rather than by a fixed CPU/GPU split. A single `vulcan.cohort` process keeps the index and a precomputed predicted-spectrum library on the GPU, while 24 CPU workers decode RAW files and 3 CPU workers run competition, protein grouping and quantification (Fig. 1b). On the closed path the GPU kernels use 0.11–0.20 GPU-seconds per run, so the critical path after setup is the host time of the GPU-stage thread, not the GPU itself.

Exact GPU search reduces CPU cost by one to two orders of magnitude

We first checked that the GPU search reproduces a CPU reference. On 3,000 random spectra each from D0 (PXD001468, two HEK runs) and D1 (PXD001077, one *Pyrococcus furiosus* run), the GPU index returned the same candidate sets, counts and per-hypothesis statistics as plain single-precision brute force. With a top-K of 500, target and decoy recall relative to exhaustive rescoring was 99.90% and 99.78% on D0 and 99.99% and 99.91% on D1. Under the same hyperscore and the same estimator, Vulcan and Sage[5] gave 19,601 and 19,605 peptidoforms at 1% FDR on D0, and 5,758 each on D1.

On D0, a complete single-process search, including Python import, CUDA initialization and index loading (excluding index build and RAW ingest), took 8.59 s and 6.0 CPU core-seconds (core-s) in the production configuration. Sage at 32 threads took 29.9 s and 237 core-s, and at 4 threads 46.5 s and 137 core-s. This corresponds to about 40-fold and 23-fold fewer CPU core-s for the search stage, respectively (CPU only; GPU time excluded). An earlier derived estimate (process total minus warm passes; 7.5 s, 4.9 core-s) gave 48-fold and 28-fold. The search phase alone took 0.24 s against Sage's 2.98 s. Throughout this paper, core-s count user plus system CPU time and exclude GPU time, and all comparator timings are for CPU-only tools on the same 32 cores. A per-run comparison therefore favours a node that has a GPU.

The reduction in CPU cost held across workflows, with larger factors where the comparator's search space grows (Fig. 2a). RAW to protein groups cost 9.1 core-s per run in Vulcan (with the opt-in speed-ups described below), against 70–120 core-s for a Sage search from mzML at 4–32 threads (8–13-fold) and a further 250 core-s for RAW to mzML conversion. Label-free quantification cost 15.3 against 266 core-s (17-fold), although these two figures come from different instruments and datasets. Phosphopeptide search with site localization and false localization rate (FLR) estimation cost 9.8 core-s per run, while Sage's phospho search, including database build and without localization, cost 3,087 core-s for two runs (158-fold). Open search over [−150, +500] Da used about 66-fold fewer core-s than Sage's open mode at 32 threads (17.5 versus 1,153 core-s for two runs), semi-specific search for limited proteolysis (LiP) 492-fold, and IT-CID/TMT search and scoring about 59-fold per fraction (0.91 versus about 54 core-s; Sage at 16 threads without the bench lock, so indicative). On the DSS cross-linking library, Vulcan's search stage alone used 1.26×10^5-fold fewer core-s than a complete xiSEARCH run (about 5×10^4-fold for search plus site localization, and about 8,900-fold less wall time for the search stage), and the whole pipeline took about 460-fold less wall time; this xiSEARCH run was timed under the bench lock (the larger DSSO comparison rests on a subset extrapolation on a contended node). Direct RAW decoding cost 17.8 core-s per 10^5 MS2 spectra on an earlier 4-core node (19.2 on the node of record) and 6.5 in a persistent worker, against 531 for ThermoRawFileParser conversion to mzML[19] on the earlier node; that comparator figure was inflated by cold network-file-system reads, and ThermoRawFileParser re-measured on the node of record (different file) cost 235 core-s per 10^5 MS2 spectra.

Cohort throughput and an estimate for one terabyte

In 96-run cohorts read from local disk, the default closed pipeline from RAW to protein groups ran at 4,570 runs per hour and 9.9 core-s per run (Fig. 2b). Exact opt-in changes to process start-up and GPU-stage threading raised this to 5,753 runs per hour at 9.1 core-s per run. These changes left all compared output tables (288 or 480 per comparison) bitwise identical, but they have not been merged into the frozen pipeline. The same changes raised label-free quantification

without match-between-runs (MBR) from 561 to 1,280 runs per hour, with MBR from 572 to 1,194, chimeric search from 2,567 to 3,867, and phospho search with localization and FLR from 2,362 to 3,172. Absolute rates varied by about ±25% between rounds on the shared node, so we compare only within a row.

From these per-run costs we estimated the wall time to quantify 1 TB of RAW data, about 1,000 runs of about 1 GB, on one machine (Fig. 2c). Vulcan would need about 0.8–1 h, and Sage with RAW to mzML conversion on the same 32 cores about 4.5–7 h. This is an extrapolation, not a measured end-to-end run. It favours a GPU node, mixes timings from different datasets, and ignores network I/O. Streaming the 1,000-run D8 cohort from PRIDE was download-bound at about 580 runs per hour, so I/O, not compute, would limit both tools in that setting.

Identification sensitivity matches the strongest comparator

For standard DDA on Orbitrap higher-energy collisional dissociation (OT-HCD) data, the frozen default scorer (v1) is a small gradient-boosted model over label-blind search features and predicted fragment-intensity and retention-time features from AlphaPeptDeep[11]. It was trained on 12 runs from 12 public projects that do not overlap any evaluation set, and it is never fitted on the data it scores. Because the main gain of such scorers comes from predicted features, the pre-registered comparator was Sage with the same AlphaPeptDeep features rescored by mokapot[16], rather than Sage alone. We compared methods at matched entrapment-estimated false discovery proportion ($\widehat{\text{FDP}}$) of 1%. Each method's own threshold was moved until its paired-entrapment estimate reached 1%, so differences in the methods' own FDR estimates do not affect the ratio.

At matched $\widehat{\text{FDP}}$, v1 gave 0.990 times the comparator's peptidoforms on D0 (95% CI 0.989–0.992) and 0.998 times on D1 (0.997–1.000; Fig. 3). Both lie above the pre-registered non-inferiority margin of 0.97. D0 and D1 were used to select v1, so these are selection-set results. Against plain Sage the gain at 1% was 2.8%, and on ten development runs held out from scorer selection (D8h; not a sealed test set) v1 gave 1.55 times as many peptidoforms per run as the hyperscore at each method's own 1% q-value (geometric mean over 9 evaluable runs, 1.41–1.71). These results suggest that the identification gain over plain Sage comes from the predicted features, and that Vulcan's scorer is at parity with a CPU pipeline that uses the same features.

For IT-CID and TMT data, a separate linear scorer with Prosit[10] features served through Koina[20] gave 1.034 (1.030–1.037) and 1.093 (1.088–1.097) times the sequences of Sage with Prosit features and mokapot on D4 and D10cf. A cross-encoder research arm (ce_s, 1.66 million parameters) gave 0.983 (0.968–0.999) times v1's identifications on D8h and failed its pre-registered rule for becoming the default.

Chimeric search treats each co-isolated MS1 feature as a separate query and applies a CONGA-style neighbour filter[21] and target–decoy competition within chimeric-rank strata (v2_sep). Against a fair comparator, Sage in chimera mode with predicted features and mokapot, v2_sep gave 1.149 (1.147–1.152), 1.137 (1.120–1.140) and 1.150 (1.069–1.203) times the peptidoforms on D0, D1 and D8h (Fig. 3). Protein groups on D8h (development data) did not increase (0.999, 0.951–1.039), and at 0.1% the peptidoform gain on D8h shrank to 1.096. The chimeric gain is therefore at the peptide level, not in protein depth. v2_sep remains a candidate rather than the default, because it has never been tested on null spectra.

Entrapment estimates stay near nominal per run and at repository scale

Because no null-spectrum gate was passed (see below), the error-rate statements in this paper rest on entrapment estimates on development data. We used two arms: paired shuffled entrapment (arm P) and natural foreign entrapment from *Arabidopsis* (arm C)[18]. The paired estimator overstates FDP; in simulation, a correctly calibrated list with true FDP 1.04% gives a mean estimate of 1.24%. A list just above 1% is therefore labelled "inconclusive" under the pre-registered rule, not "violated".

At nominal 1%, per-run arm-P $\widehat{\text{FDP}}$ for v1 was 1.01% on D0 (95% CI 0.94–1.07%), 0.87% on D1 (0.73–1.01%) and 1.02% on D8h (0.98–1.06%; Fig. 4a). For the comparator with predicted features and mokapot, it was 1.05%, 0.90% and 0.97%. No list from v1, v2_sep or this comparator was classed as "violated" on any development set. Within chimeric strata, the pre-registered Group-walk procedure[22] was rejected because its rank ≥3 stratum reached 3.09% (2.38–3.79%) on D0, a "violated" verdict (Fig. 4c). Per-stratum competition kept every stratum at 1.02–1.15% on D0 and D8h. Sage's chimera mode, in contrast, accepted rank ≥2 identifications at 9.42% $\widehat{\text{FDP}}$ on D0 (not evaluable, too few targets) and 5.34% (4.70–5.99%) on D8h ("violated"). The chimeric ratios above therefore favour the comparator.

At repository scale we streamed 1,000 human OT-HCD runs from 142 PRIDE projects (D8; 982 runs from 140 projects completed) and pooled the best evidence per peptidoform into one global paired competition (L6). At N = 1,000 the global list had arm-P peptide $\widehat{\text{FDP}}$ of 1.16% and protein-group $\widehat{\text{FDP}}$ of 1.00% (Fig. 4b), and about 358,000–360,000 peptidoforms

were accepted (358,164 in the arm-P evaluation table; 360,392 in the master-list example of the same results notes). A naive union of per-run 1% lists reached 9.3% at the peptide level and 19.1% at the protein level. This pattern is consistent with earlier reports that merged per-run lists inflate error[4,14]. The 0.16-point excess of the global list has two identified sources: an entrapment-share effect on arm P and a target-favouring tilt of the natural arm C (arm C was 1.21%). Only one entrapment and one decoy seed were used at this scale.

PTM localization and label-free quantification

Site localization enumerates all phosphosite configurations, including decoy configurations on alanine (primary) or leucine (secondary) residues[23], and estimates FLR from the rate at which decoy residues are chosen. On synthetic-library truth from PXD000138[24] and PXD009449, the true FLR of accepted mono-phospho sites at a nominal Ala-decoy FLR of 1% was 0.29% and 0.86%, and at 5% it was 3.43% and 4.44% (Fig. 5a). Point estimates were thus calibrated or conservative. A stricter pre-registered rule, which requires the one-sided 95% upper bound of the true FLR to be at or below the nominal level, was met only on PXD000138. On PXD009449 the result was inconclusive at both levels (upper bounds 1.34% at 1% and 6.93% at 5%), because only 183 peptides carried that set. Against pyAscore at matched true FLR, site recall was higher on PXD009449 at 1% (+2.0 points, 95% CI 1.1–2.8), not different on PXD000138 at 1%, and lower on PXD000138 at 5% (−1.6 points, −3.2 to −0.3; Fig. 5b). Localization settings were tuned on these two datasets.

Localization was not reproducible run to run. Floating-point atomic sums on the GPU break exact score ties at random, and 313 of 45,845 PSMs on PXD009449 changed their best site configuration between two identical runs. No development verdict changed over 11 repeats. A deterministic tie rule removes the variation but accepted 1.7% fewer sites at 1% FLR on PXD009449; it was not adopted, because the FLR evaluation was pre-registered with the original rule.

For label-free quantification, MS1 features are linked to accepted identifications, re-integrated over a fixed window and rolled up with MaxLFQ[25] or directLFQ[26]. On a three-species benchmark (D9; ProteoBench DDA-Astral human–yeast–*E. coli*, 6 runs, MBR off), the median absolute $\log_2$-ratio error on 13,674 peptidoforms quantified by both tools was 0.118 for Vulcan and 0.1145 for Sage (Fig. 5c). Vulcan was worse on low-abundance *E. coli* peptides (0.160 versus 0.121). Quantification choices were made on D9, so this parity is a development result. MBR is opt-in and reports a transfer q-value. Its measured false-transfer FDP was 0.63–1.23% at nominal 1% on D9 and 0.18–0.79% on an independent two-proteome set (D12)[27], but on D12 the transfer decoy under-counted false transfers about 2.4-fold. For TMT, reporter intensities equalled Sage's (r = 0.9999996).

Specialized workflows share the same index

The same resident index supports cross-linking, LiP and co-fractionation (CF) analyses (Supplementary Table 3). For the cleavable cross-linker DSSO on a synthetic ribosomal library (D3b)[28], Vulcan reported 334 heteromeric residue pairs at a true FDP of 1.2% (nominal 1%), against 459 at 2.0% for the submitter's xiSEARCH result, about 73% of its sensitivity at a lower error. On a DSS peptide library (D3)[29], Vulcan reported no false heteromeric links, but self residue pairs at nominal 5% had a true FDP of 9.2%; xiSEARCH was better on these. Homo-peptide pairs cannot be estimated by target–decoy counting[30,31] and are reported without FDR. For LiP (D7), semi-specific search gave 26,976 peptidoforms at nominal 1%, and all 20 significant trypsin-adjusted α-synuclein peptides from MSstatsLiP[32] fell in the fibril core (residues 37–99). No entrapment was run for the semi-specific class. For CF data (D10cf), fraction-local transfer placed no values three or more fractions away from MS2 evidence (off-peak), whereas naive global MBR produced 9% off-peak values but a slightly higher CORUM precision–recall area on the common protein set (0.088 versus 0.082; no confidence interval computed), so the design hypothesis was only partly supported.

Pre-registered null-spectrum gates failed or could not be evaluated

Entrapment tests whether the final list contains too many foreign sequences, but not whether a learned scorer prefers target over decoy sequences on spectra that have no correct answer. The protocol therefore required null-spectrum gates for any scorer with learned or predicted components (Fig. 6). Null spectra came from six wrong-species projects, mixed into a human run after a homolog filter, plus precursor-offset spectra. The tail-G2 gate requires that, among null spectra passing the scorer's own 1% threshold, the target fraction has an upper bound ≤ 0.60 and exceeds the hyperscore pipeline's by at most 0.10. G1 requires that the scorer does not correlate with a sequence-only target-versus-decoy classifier on nulls, and G3 that top-K retrieval treats targets and decoys alike.

The frozen v1 scorer failed all three (Amendment A1). Of 751 tail null spectra, the target fraction was 0.715 with an upper bound of 0.742, and the excess over the hyperscore had an upper bound of 0.124. The correlation with the sequence classifier was ρ = 0.048, and its 99% CI (0.019–0.076) was not contained in the required ±0.05 band. G3 failed on the

clean-null subset by one bin at 1.03 percentage points. After the failure, we found that one null project (*Staphylococcus aureus*) contained human skin proteins and that the tail offset nulls were shifted human spectra. Without these two sources, tail-G2 and G1 would pass. This explanation was found after the result, is unconfirmed, and did not change the recorded verdict.

The later attempts to re-test the scorer did not produce a gate result. A fresh null set (A2) admitted 6 projects against a pre-registered minimum of 10 and was declared not evaluable. Ion-trap nulls yielded 581 usable spectra against 120,000 needed. A third null-set design (A4) had a simulated probability of only 0.08–0.43 of reaching its minimum sample for v1 (and about 0.06–0.4 of passing a clean v1) and was not adopted or run. A human removed-peptide null test (A6) was ruled infeasible by its own pre-registered rule, because the measured admission rate gave a probability of 0.63 of reaching its minimum, against 0.8 required. Its acceptance test also showed that the cluster bootstrap behind the original G2 gate undercovers with about 37 clusters (coverage 0.923–0.944 against 0.95). The chimeric and IT-CID scorers have never been tested on null spectra.

Amendment A5 then knowingly relaxed the conditions for a single sealed look at the held-out sets T1 (a ProteoBench DDA benchmark) and T5 (a yeast run for the memorization arm), on condition that every FDR-filtered number carries the gate table. The look opened on 28 October 2026 at 14:23 CET (13:23 UTC in the sealed-run ledger). T5 completed. T1 was killed in its final scoring stage when the compute node was wiped. The attempt was closed as "failed (unterminated)" and all outputs were deleted unread. No held-out test-set comparison exists, and the FLR, cross-linking and repository test sets (T2–T4) remain sealed. A5 allows at most two retries from the same frozen commit, each re-running T5 and T1, and only one attempt has been used, so up to two retries remain allowed; a retry first needs the lost node-local mokapot environment to be rebuilt and checked on development data.

Discussion

Here, we show that a GPU/CPU co-designed engine can process DDA runs from vendor files to protein groups at a CPU cost of about 10 core-s per run on development data (GPU time excluded), roughly 30–40-fold below a CPU-only search engine plus file conversion. Several engineering choices made this possible. A long-lived process keeps the index resident, so start-up costs are paid once per cohort. Vendor files are decoded directly, so mzML conversion is avoided. Fragment matching, scoring and predicted-feature computation run as GPU kernels. Exactness was enforced throughout: bit-identity under label swaps and bitwise-equal outputs for every speed-up. Speed gains are therefore not traded for silent changes in output. At this cost, reanalysing a terabyte of RAW files would become a task of about an hour on one GPU node, an estimate that rests on per-run costs rather than a measured run.

The sensitivity results are best read as parity. The frozen scorer matched Sage with the same predicted features and mokapot within 1% of peptidoforms at matched entrapment-estimated error, and its gain over plain Sage came from those predicted features. Chimeric search added about 15% peptidoforms against a chimeric comparator, but no protein groups. These comparisons used a comparator that received the same features, which we consider the fair baseline; MSFragger and DIA-NN[33] were not run.

Global competition across runs kept entrapment-estimated peptide and protein error near 1% up to 1,000 runs, whereas the naive union of per-run lists reached 9.3% and 19.1%. This supports the use of a global analyte-level procedure for repository-scale resources[14,34]. The result holds for an entrapment estimate with one seed on development data. It is not a test of whether the scorer itself is null-symmetric.

The main limitation of this work is its statistical status. The frozen scorer failed the only completed null-spectrum test. Our post hoc explanation, contamination of the null material, is plausible but unconfirmed. Three further attempts to test it ended as not evaluable, not run or infeasible, for reasons of power that we could have anticipated earlier. All accuracy and error-rate numbers come from development sets, and several of those sets were used to select or tune the methods (D0 and D1 for the scorer, PXD000138 and PXD009449 for localization, D9 for quantification). The sealed held-out evaluation was interrupted by a node failure, and its outputs were not seen. We therefore cannot state that Vulcan controls FDR at its nominal level, only that entrapment estimates on development data stayed near 1% and were never classed as violated. Running the sealed T1 look on a stable node, re-testing the scorer with a null set of adequate power, and replacing the undercovering bootstrap would address these gaps.

Other limitations are more specific. Localization was not reproducible for about 0.7% of PSMs on PXD009449, and a deterministic rule costs sensitivity. Cross-linking sensitivity was below xiSEARCH. The MBR transfer decoy undercounted false transfers on one dataset. Quantification was at parity, not better, and worse for low-abundance proteins. All timings come from a shared node with load averages of 14–110, some comparator timings were taken without the

bench lock, and several reported throughputs use opt-in paths that are not yet merged. The ion-trap, Astral, Bruker and data-independent acquisition (DIA) settings remain untested or outside the default scorer's instrument class.

Several approaches did not work and are reported as negative results. A bi-encoder retrieval stage reached 79.5% recall at 1,000 candidates in open search and was no faster than the inverted index. A cross-encoder reranker did not beat the static scorer. Nineteen additional label-blind features gave no gain on held-out training folds. Per-run score calibration cost 13–72% of identifications. Natural foreign null sequences beat reversed decoys in 54.4% of competitions under the frozen score, against 51.6% under the hyperscore, which suggests that decoy realism deserves further study.

We release the engine, the pre-registration history and the gate record together, so that the speed and architecture can be reused and the outstanding evaluation can be completed by others. The pre-registration served its purpose here: it turned an unexpected failure into a recorded outcome rather than a revised analysis. This account may be useful for other groups building learned scorers for repository-scale reanalysis.

Methods

Hardware, timing and reporting conventions

Except for the first ingest benchmark, which ran on an earlier 4-core node with an RTX 3090 GPU, all measurements reported here used one node with two AMD EPYC 9354 processors (32 cores in the process affinity mask, simultaneous multithreading off), one NVIDIA A100 80 GB PCIe GPU and 1.5 TB of RAM. The node was shared with other jobs; the load average was 14–110 across measurements and is recorded with each timing. Timed runs used a file lock to serialize benchmarks where stated. Several comparator timings (the xiSEARCH DSSO subset run, Sage on D4) were taken without the lock or on a contended node and are indicative only. Every timing record carries a hardware fingerprint. Core-s are user plus system CPU seconds measured by GNU time or process accounting and exclude GPU time, which is reported separately as GPU-seconds where available. Comparators ran on the same 32 cores without a GPU; Sage was also timed at 4 threads. Runs per hour are RAW to the stated output in a `vulcan.cohort` process, from local disk with a pre-warmed page cache unless stated. Throughputs vary by about ±25% between rounds on the shared node and are compared only within a session. Thread pools are sized from the CPU affinity mask, not from the reported CPU count.

Ingest and spectral store

Thermo RAW files are decoded in-process with the vendor RawFileReader library in supervised persistent worker processes, after staging the RAW file to local NVMe storage (memory-mapping files on the Lustre file system stalled). All MS1 and MS2 centroids are written to an Arrow IPC store with spectrum metadata (precursor, isolation window, activation, analyser, normalized collision energy, scan filter). Peak preprocessing (top-k peaks, deisotoping, precursor removal) is computed at search time on the GPU and never requires re-ingest. TMT reporter peaks are stored separately before any peak filtering.

Proteome index and decoys

Digestion and variable-modification enumeration use the sage-core library through a Python binding. Decoys are pseudo-reversed sequences with the C-terminal residue fixed and I/L treated as equivalent; a target–decoy collision is replaced by a shuffle seeded by a hash of the peptide and seed. Each record stores its paired partner. Entrapment sequences receive their own paired decoys. Peptidoform identifiers are xxhash64 values of the canonical peptidoform, so identifiers do not depend on storage order, and ties are broken by hashes. Labels are stored only in a separate table read by the statistics layer. The GPU index is a compressed sparse row structure over (log-ppm fragment bin × chunk of 8,192 peptidoforms) with 4 bytes per posting. For D0 it held 175,429,652 postings, equal to Sage's fragment count, used 1.5 GB of GPU memory and built in 0.53 s. The phospho index holds 184.5 million peptidoforms (34 GB).

Candidate generation and scoring kernels

The query unit is a precursor hypothesis: one row per spectrum, charge, isotope error and, in chimeric mode, co-isolated MS1 feature. Kernel K1 assigns one warp per (hypothesis, chunk tile), counts matched fragments in shared memory, applies the same threshold to every candidate regardless of label and keeps the top K = 500 candidates per spectrum. Each shortlist is closed under target–decoy pairing, so a scored target always has its paired decoy scored with the same features. Kernel K2 reproduces Sage's candidate scoring in single precision and computes label-blind features (hyperscore, matched ion counts and intensities, ion-ladder lengths, mass-error statistics, score differences to the next candidate regardless of label). Predicted features are fragment-intensity similarities (spectral angle, Pearson correlation,

entropy similarity, observed fractions) and retention-time residuals from AlphaPeptDeep[11] (peptdeep 1.5.1). A library covering the human production index (7.06 million peptidoforms) and the *P. furiosus* index at charges 2–4 and all six collision-energy grid values is precomputed once (24.9 GB, mirrored in GPU memory); other peptidoforms are predicted lazily and cached by content hash. Collision energy is chosen per run from a grid of six values between 20 and 35 by the median spectral angle of a label-blind set of high-scoring PSMs. The predicted retention time itself is excluded from the model, because it separates targets from decoys on sequence alone.

Scorers

The frozen OT-HCD scorer v1 (`static_gbm_pred`) is a histogram gradient-boosted classifier (depth 3, 8 leaves, minimum leaf size 200) over the K2 and predicted features. It was trained on 12 runs from 12 PRIDE projects (human, mouse and *E. coli*; 829,000 PSMs), selected with seed 11 from a crawl frame after excluding every development, sealed and held-out accession, predictor training sources, yeast and *Arabidopsis*. Training labels were derived by iterative target–decoy competition on the training runs only, with hyperparameters chosen by leave-one-project-fold-out validation. The model was then refitted on all folds and frozen by hash before evaluation. The chimeric scorer for v2_sep is a linear model over the v1 feature set retrained on 40 runs from 40 projects. The IT-CID scorer is a linear model over label-blind features and Prosit[10] CID/TMT predictions obtained through a pinned local copy of Koina[20] responses, because Koina predictions are not bit-deterministic between rebuilds.

Target–decoy statistics

All hypotheses of a spectrum compete first. Then PSM-level target–decoy competition is followed by peptide-level competition between each target and its own paired decoy (paired double competition[35]). FDR is estimated as (D + 1)/T and made monotone. In chimeric search, each co-isolated MS1 feature group forms its own query unit; a winner that shares more than 5% of its b/y ions (index ≥ 3, 0.02 Da bins) with a higher-scoring winner of the same spectrum is removed[21]; and competition is run separately within chimeric-rank strata {1, 2, ≥3} (v2_sep). Group-walk[22] over the same strata was pre-registered and rejected (see Results). Protein groups use picked protein-group FDR[14] with the best peptide score. At repository scale (L6), the best evidence per peptidoform across runs is kept per side, pooled into a global paired competition and then into picked protein groups. Each run's list carries both a run-level and a global q-value. The estimators were checked by Monte-Carlo simulation (≥10^4 replicates).

Entrapment and decision rule

Arm P uses paired shuffled entrapment at ratio r = 1 with its own decoys[18], $\widehat{\mathrm{FDP}} = (N_E + N\{\mathrm{E}\geq\mathrm{s}>\mathrm{T}\} + 2\cdot N_\{\mathrm{E}>\mathrm{T}\geq\mathrm{s}\})/(N_T + N_E)$. Arm C uses the *Arabidopsis* proteome, filtered for I/L-identical, single-substitution and isobaric neighbours of human peptides, with $\widehat{\mathrm{FDP}}_C = N_E(1 + 1/r)/(N_T + N_E)$. Arm M searches a yeast sample against yeast plus human entrapment, to test memorization by sequence-aware models. The statistic is the mean $\widehat{\mathrm{FDP}}$ over replicates (3 entrapment seeds × 5 decoy seeds on development data; fixed seeds 2001–2003 and 1001–1005), with a 95% CI from bootstrapping target–entrapment pairs plus between-seed variance. A list is "supported" if the mean is ≤ α and the upper CI ≤ 1.5α, "violated" if the lower CI of the lower-bound estimator $N_E/(N_T + N_E)$ exceeds α, "not evaluable" if $\alpha\cdot N_T < 20$, and "inconclusive" otherwise. Sealed entrapment seeds (2004–2006) were never used.

Comparisons at matched $\widehat{\mathrm{FDP}}$

For the evaluation driver, a single multi-replicate index defined 25 replicate databases (5 production, 15 arm P, 5 arm C), and every method searched exactly these peptide sets. Comparators were Sage 0.14.7 (with retention-time prediction on), Sage with mokapot 0.10.0, Comet 2026.01 with mokapot, and Sage with Vulcan's nine AlphaPeptDeep and retention-time features appended and rescored by mokapot (the strongest comparator). For chimeric search the comparator was Sage chimera mode (R = 2 or 5 co-isolated candidates, chosen by a pre-registered rule) with the same features and mokapot. Because mokapot 0.10 ignores its seed argument, it was run through a seeding wrapper. Units were identical across methods (peptidoform: I/L-collapsed sequence with methionine oxidation positions). The ratio of identifications was computed at the first crossing of arm-P $\widehat{\mathrm{FDP}}$ = 1% on a 160-point grid of each method's own q-value, with run-level counts summed over runs and averaged over decoy shuffles, and a 95% percentile bootstrap (2,000 draws, seed 11) over runs × decoy shuffles. The pre-registered non-inferiority margin was 0.97.

Null-spectrum gates

Null spectra came from six OT-HCD projects of non-human species (*P. furiosus*, yeast, *Trypanosoma brucei*, *S. aureus*, *Synechococcus* PCC 7002 and *Bacillus cereus*) and from precursor offsets of n + 0.5 Da. In total 894,579 species-null

spectra were quantile-mapped onto the host retention-time gradient and mixed with a D0 human run in 45 mixtures of 20,000. A homolog filter removed any null whose matched peptide or its pair partner shared a 6-mer with, or was within Hamming distance 2 of, the source proteome or contaminants, leaving 81,390 clean nulls. Tail-G2, G1, G3, G4 and G5 were evaluated as defined in the Results and in the pre-registration (DESIGN §6.4). The protocol versions and their amendments (A1–A6) are tagged or committed in the repository and summarized in Supplementary Table 2.

Pre-registration and sealed evaluation

The statistical protocol was frozen at a git tag before any sealed file was accessed, and every amendment was dated before the affected results were seen. Sealed raw files were stored with no read permission and could be unlocked only by a script that requires a clean working tree, a milestone tag and a ledger entry written before the run. The single A5 look was opened from commit 4fe2e2a with tag `milestone-M7-2026-10-28`. The ledger records T5 as completed and T1 as "failed (unterminated)", with outputs deleted unread.

Localization and FLR

For each accepted phosphopeptidoform, all site configurations are enumerated, including configurations that place the modification on Ala (primary) or Leu (secondary) decoy residues, and site-determining ions are scored without reference to residue identity. FLR is estimated as n_pA·(N_STY/N_A)/n_pSTY. Truth was defined before any localization result from the synthetic design of PXD000138 pools and the published pool lists of PXD009449. Pools with positional isomers were excluded. Site recall was compared with pyAscore (documented defaults) at matched true FLR with a paired sequence-cluster bootstrap.

Quantification

Accepted PSMs are linked to MS1 isotope-envelope features from a batched feature finder (87% of PSMs linked on D9). Intensities are re-integrated over a window of fixed width in every run, aligned across runs and rolled up with MaxLFQ[25] or directLFQ[26]. MBR is opt-in. Each transfer carries a transfer q-value from decoy peptides and retention-time- and m/z-shifted decoy peaks, and transfers are not counted as identifications. TMT reporters are corrected for isotope impurity. Accuracy on D9 is the median absolute error of $\log_2$ ratios against the known species ratios on peptidoforms quantified by both tools.

Cross-linking, LiP and co-fractionation

For DSSO, doublets separated by 31.972/z are detected on the GPU, and candidate peptide masses are searched with the alkene and thiol stubs as mass shifts. For DSS, an α-first open search is followed by a mass lookup of the β peptide. The cross-link spectrum match score is the minimum of the two peptides' evidence, and linear, mono-link and loop-link explanations compete in the same search. FDR is estimated as (TD + 1 − DD)/TT at the residue-pair or protein–protein level, separately for self and heteromeric links[30,31]. LiP search is semi-specific with class-matched decoys, and its output is passed to MSstatsLiP[32]. For CF data, transfers are restricted to neighbouring fractions.

Datasets

Development sets were D0 (PXD001468, 2 runs), D1 (PXD001077, 1 run), D2 (development pools of PXD000138[24] and PXD009449, plus PXD007058[36]), D3 (PXD014337)[29], D3b (one PXD029252 file)[28], D4 (PXD016766, 12 TMT11 fractions), D7 (PXD031616), D8 (1,000 human OT-HCD runs from 142 PRIDE projects, selected with seed 11 and streamed; D8h is its first 10 runs from 10 projects), D9 (ProteoBench DDA-Astral human–yeast–*E. coli*, 6 runs), D10cf (PXD015406), D11 (PXD003881) and D12 (PXD014415)[27]. Sealed sets were T1 (ProteoBench DDA-QExactive), T2 (held-out PXD000138 and PXD009449 pools), T3 (a separate PXD029252 file), T4 (a stratified PRIDE sample) and T5 (PXD052597, two yeast runs). Training, null and evaluation accessions were checked to be disjoint, and against the published training sets of the third-party predictors. Sequence databases were UniProt release 2026_03.

Statistics and reproducibility

All confidence intervals are 95% percentile bootstrap intervals unless stated. Seeds were fixed in the pre-registration. Every reported number is linked to a results directory containing its configuration, commit hash and metrics file, and every plotted value is mapped to its source key by a figure script.

Data availability

All raw data used here are public. Development and training data are available from PRIDE under the accessions listed in Methods and in Supplementary Table 1, and from ProteoBench. The repository contains per-file manifests, checksums and fetch scripts. Derived tables from the 1,000-run D8 stream were not retained. No held-out (sealed) result exists.

Code availability

Vulcan's source code, pre-registration (tags `prereg-v0.4` to `prereg-v0.4.4`), amendments and reviews, the sealed-run ledger, all results directories and the figure script are kept in the project's git repository. The final freeze commit is 06ce624 and the sealed-look commit is 4fe2e2a. The speed-ups described as opt-in are on an unmerged branch (commit 7f64a72). The repository has not yet been deposited in a public archive.

AI involvement statement

The engine, experiments, analyses, figures and this manuscript were produced by a team of AI agents (Claude Opus 5.5 and Claude Sonnet 5.5). The agents used a shared compute node and a written pre-registration, and they wrote independent internal reviews of design documents and amendments. No human expert has reviewed or verified the code, the results or this text.

Competing interests

None declared by the AI team.

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Figure legends

All values come from development datasets; no held-out test-set result exists. Timings come from a shared node (2× AMD EPYC 9354, 32 affinity cores, plus one NVIDIA A100 80 GB; load average 14–110). Core-s are user plus system CPU seconds and exclude GPU time. "Nominal" is the requested FDR threshold. $\widehat{\text{FDP}}$ is the false discovery proportion estimated by entrapment (arm P, paired shuffled; arm C, natural foreign *Arabidopsis*). "Verified" denotes an FDP measured against a ground-truth design. No FDR claim has passed the pre-registered null-spectrum gates (Fig. 6).

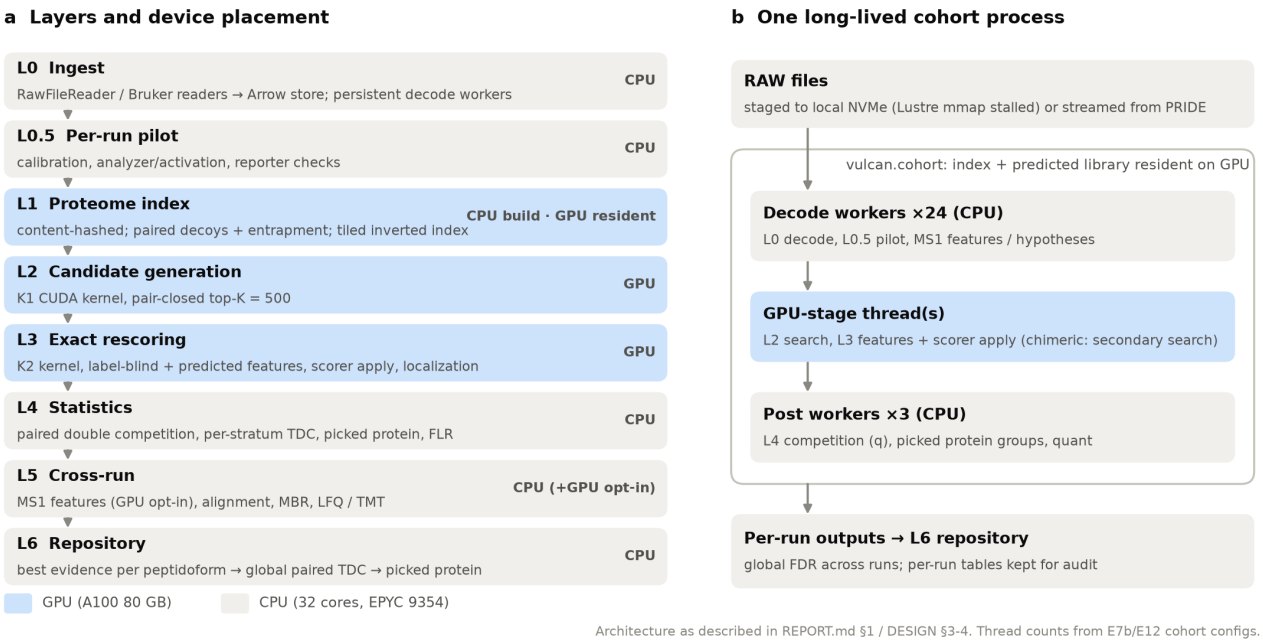


Figure 1 | Vulcan architecture and CPU/GPU placement. **a**, Processing layers L0–L6, coloured by the device that runs them in production (blue, GPU; grey, CPU). The proteome index (L1) is built on the CPU and kept resident on the GPU. Candidate generation (L2) and exact rescoring with label-blind and predicted features (L3) run on the GPU. Statistics, cross-run quantification and repository-scale FDR (L4–L6) run on the CPU; GPU MS1 feature finding (L5) is an exact opt-in path. A label-swap and permutation test (G0) checks that L2 and L3 outputs do not depend on target or decoy labels. **b**, A single long-lived `vulcan.cohort` process keeps the index and the predicted-spectrum library on the GPU. Work flows through 24 CPU decode workers (RAW decode, per-run pilot, MS1 features), one or two GPU-stage threads (search and scoring) and 3 CPU post-processing workers (competition, picked protein groups, quantification). Per-run outputs feed the L6 global FDR.

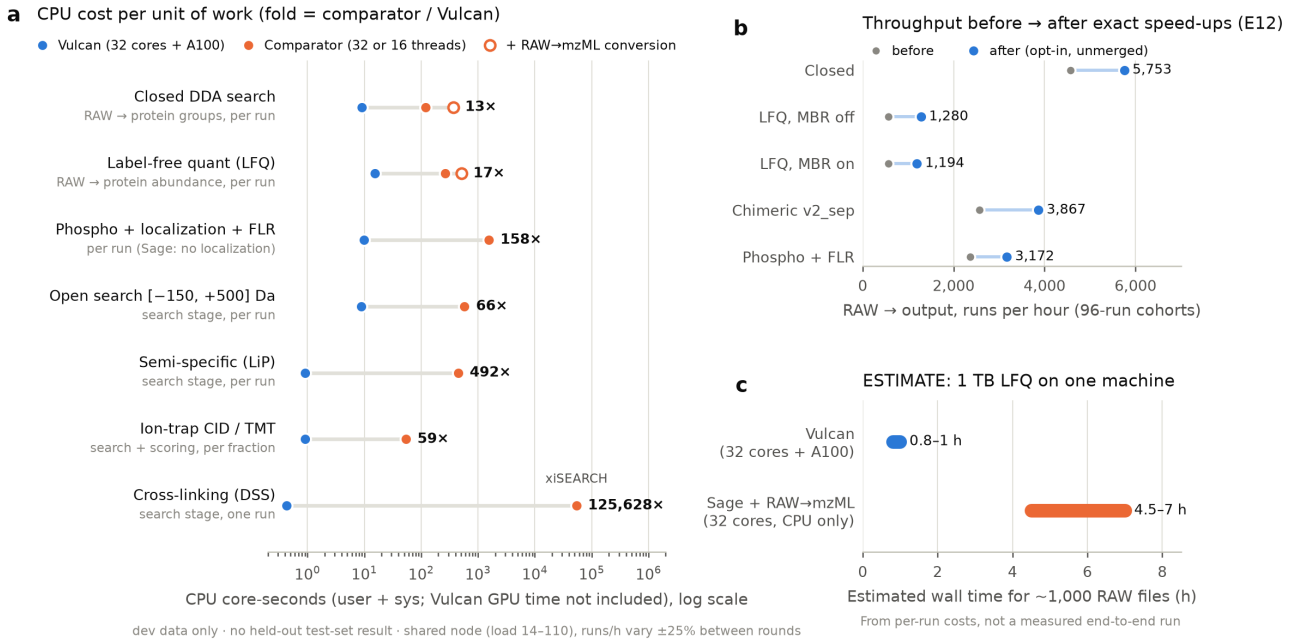


Figure 2 | Compute cost and throughput. **a**, CPU core-s per unit of work for Vulcan (blue; 32 cores plus A100) and a CPU-only comparator (orange), log scale. Fold labels are comparator core-s divided by Vulcan core-s; GPU time is not included for Vulcan. Rows: closed DDA, RAW to protein groups (Vulcan) versus Sage 0.14 search from mzML at 32 threads (hollow circle adds 250 core-s for RAW to mzML conversion); label-free quantification, RAW to protein abundance with MBR off, versus Sage search plus LFQ on Astral D9 (not like-for-like data); phospho search with localization and Ala-decoy FLR versus Sage phospho including database build, without localization; open, semi-specific (LiP) and cross-linking search stage only; IT-CID/TMT search plus scoring per fraction versus Sage at 16 threads, unlocked. Cross-linking (DSS) is compared with xiSEARCH 1.8.12 at 32 threads under the bench lock; the 1.26×10^5-fold ratio

sets Vulcan's search stage against the full xiSEARCH run, and the whole-pipeline wall-time ratio is about 460-fold. **b**, RAW-to-output throughput of 96-run cohorts before and after exact opt-in speed-ups. These speed-ups leave outputs bitwise identical but are not merged into the frozen path. Runs per hour varied by about ±25% between rounds, so compare within a row. LFQ and phospho rows assume the experiment-level step runs serially after the cohort, which is conservative. **c**, Estimate, not a measured run: wall time to quantify 1 TB of RAW files (about 1,000 runs of about 1 GB) on one machine, from per-run costs. Vulcan, 0.8–1 h, from 1,194–1,280 runs per hour measured on 96-run cohorts from local disk; Sage with conversion, 4.5–7 h, the lower bound assuming ideal 32-core scaling. The estimate favours a GPU node, mixes timings from different datasets and ignores network or repository I/O (streaming from PRIDE was download-bound at about 580 runs per hour). MSFragger and DIA-NN were not run.

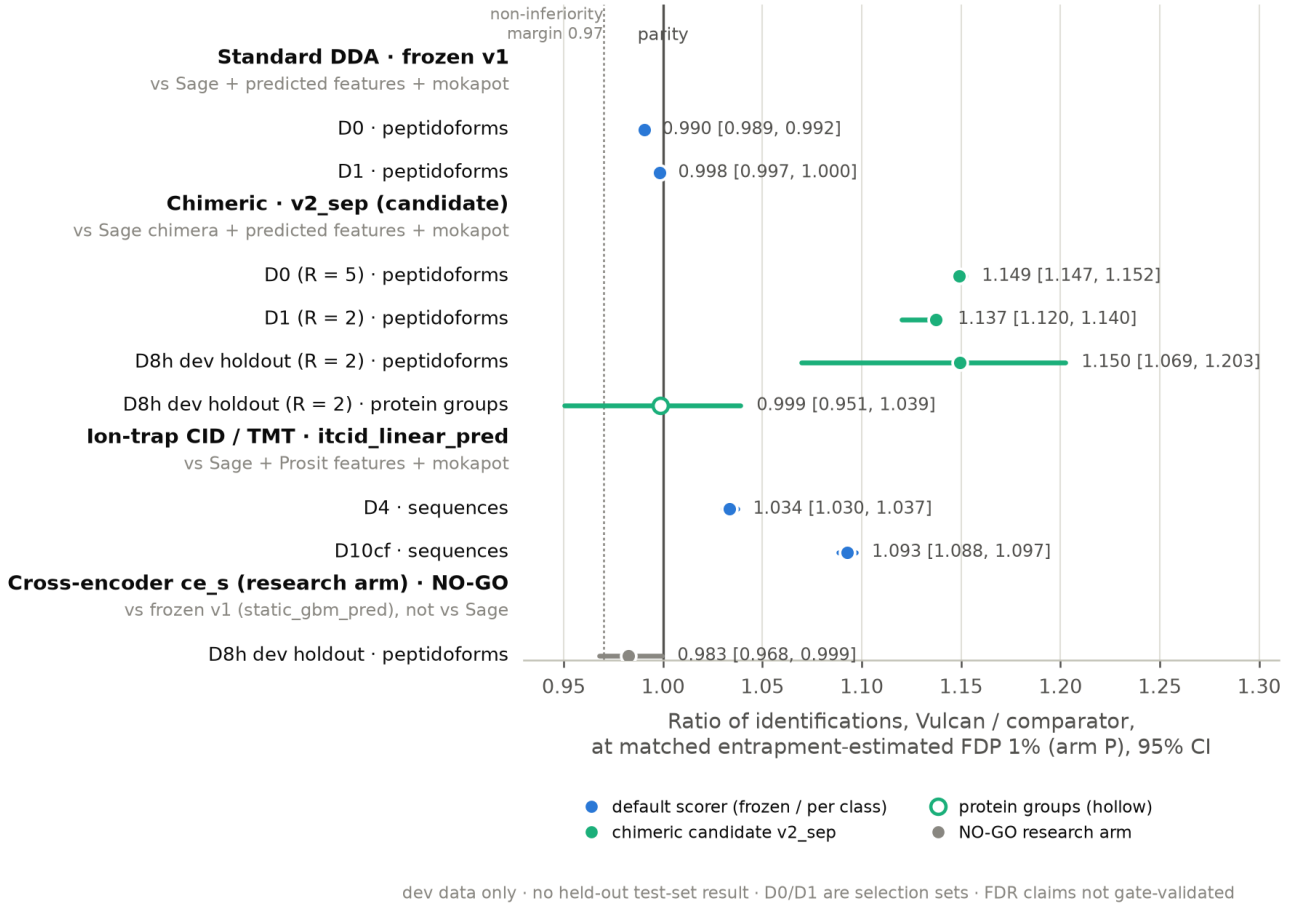
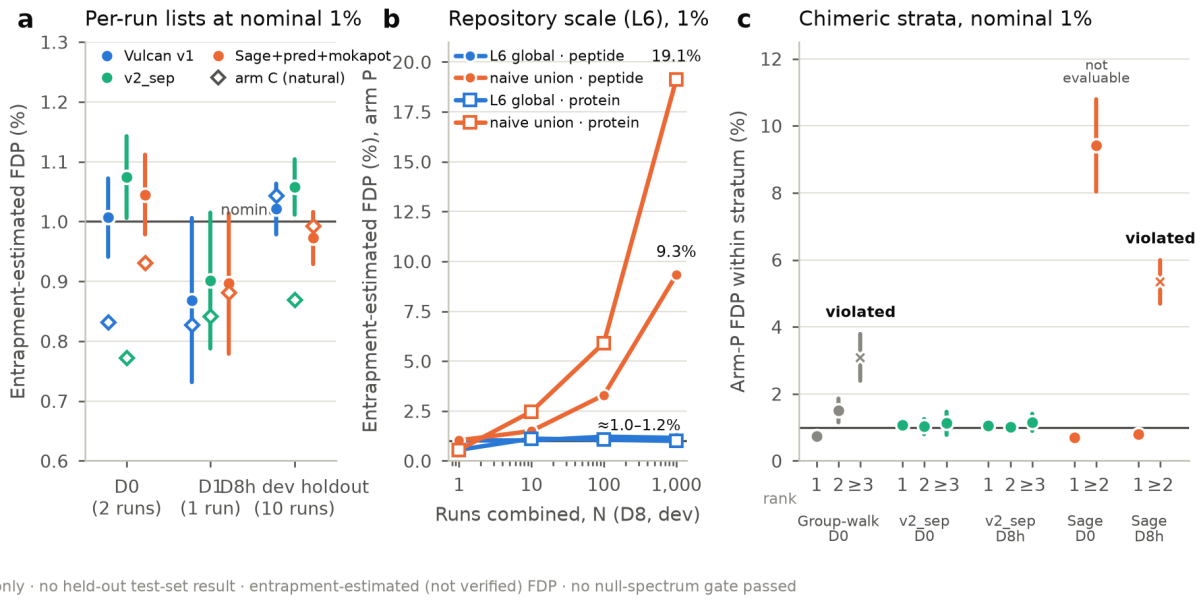
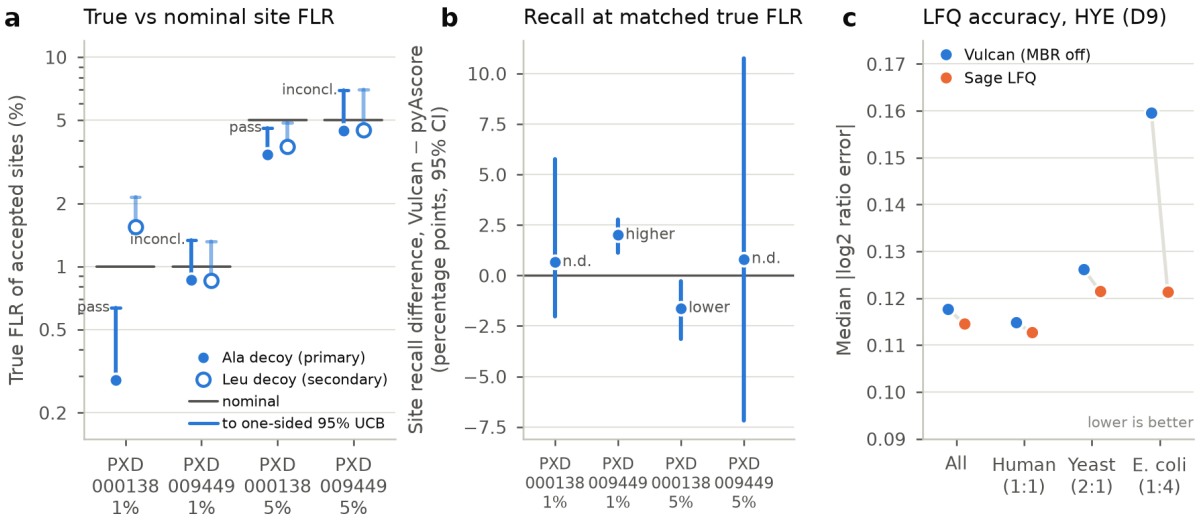


Figure 3 | Identification sensitivity relative to the strongest comparators. Each point is the ratio of Vulcan identifications to comparator identifications at matched arm-P FDP̂ of 1%. Each method's own q threshold is moved until its FDP̂ reaches 1%; identifications are summed over runs and averaged over 5 decoy shuffles. Bars, 95% percentile bootstrap CI over runs × decoy shuffles (runs only for IT-CID; D1 has a single run). Solid line, parity; dotted line, pre-registered non-inferiority margin (0.97). Standard DDA: frozen v1 versus Sage with the same AlphaPeptDeep features and seeded mokapot, on D0 and D1 (selection sets), peptidoforms. Chimeric: candidate v2_sep versus Sage chimera mode with predicted features and mokapot (R = 5 on D0; R = 2 on D1 and the D8h development holdout), peptidoforms; hollow, protein groups on D8h. IT-CID/TMT: `itcid_linear_pred` versus Sage with Prosit features and mokapot on D4 and D10cf, sequences. Grey: research cross-encoder ce_s versus frozen v1 (not Sage) on D8h, which failed its pre-registered rule. Sage chimera rank ≥2 identifications exceed nominal FDP̂ (Fig. 4c), so chimeric ratios favour the comparator.



dev data only · no held-out test-set result · entrapment-estimated (not verified) FDP · no null-spectrum gate passed

Figure 4 | Error control as estimated by entrapment (no null-spectrum gate passed). **a**, Per-run peptidoform lists at nominal 1%: arm-P FDP̂ with 95% CI (filled circles) and arm-C FDP̂ (hollow diamonds), for frozen v1 (blue), v2_sep (aqua) and Sage with predicted features and mokapot (orange) on D0, D1 and D8h. The paired estimator overstates FDP by about 20%, so values just above 1% read as "inconclusive" under the pre-registered rule; no list is "violated". **b**, Repository-scale FDR on up to 1,000 runs (D8; 982 completed, 140 projects), frozen v1 score, arm P, nominal 1%: L6 global procedure (blue; best evidence per peptidoform, global paired competition, picked protein groups) versus a naive union of per-run 1% lists (orange). Circles, peptide level; squares, protein level. One entrapment and one decoy seed. **c**, Arm-P FDP̂ within chimeric-rank strata at nominal 1%, with 95% CI; crosses mark "violated" strata. Group-walk over strata (D0; rejected because stratum ≥3 was violated), v2_sep per-stratum competition (D0, D8h; no stratum violated) and the Sage chimera comparator by Sage rank (R = 5 on D0, R = 2 on D8h). Vulcan strata are evaluated at the matched-FDP̂ threshold, Sage ranks at Sage's own q ≤ 1%. Sage rank ≥2 read-outs were pre-registered as descriptive only. v1 failed the only completed null-spectrum test (A1), and no null-spectrum test of v2_sep exists.



dev data only · no held-out test-set result · PXD000138/PXD009449 settings tuned on dev · T2 sealed, not run

Figure 5 | PTM localization and label-free quantification. **a**, Verified false localization rate of accepted mono-phospho sites on synthetic-library truth (PXD000138, PXD009449) at nominal FLR thresholds set by the Ala decoy (filled, primary) or Leu decoy (hollow, secondary). Horizontal bars, nominal level; whiskers, one-sided 95% upper confidence bound, which the stricter pre-registered rule requires to be at or below nominal. Labels give the Ala verdict under that rule; the rule is met only on PXD000138, and PXD009449 is inconclusive at 1% and 5% (183 peptides). **b**, Difference in site recall, Vulcan minus pyAscore (documented defaults), at matched true FLR with the Ala decoy; 95% CI from a paired sequence-cluster bootstrap; n.d., no difference shown. Localization settings were tuned on these datasets; localization is not run-to-run reproducible for about 0.7% of PSMs (GPU atomic ties); values come from one dry run of the pre-registered

FLR driver, and verdicts were stable over 11 repeats. The sealed FLR test set was not run. **c**, Label-free quantification accuracy on the D9 Astral human–yeast–*E. coli* benchmark (6 runs, MBR off): median absolute $\log_2$-ratio error against expected ratios on the 13,674 peptidoforms quantified by both Vulcan (blue) and Sage (orange), overall and per species. Quantification choices were made on D9.

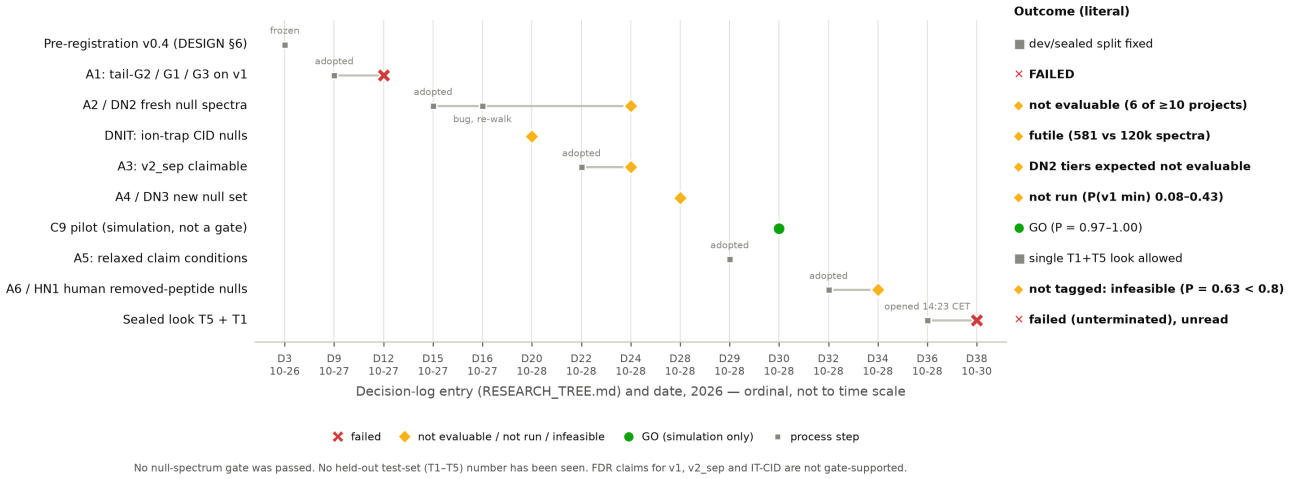


Figure 6 | History of the pre-registered statistical gates. The x-axis lists decision-log entries in order (ordinal spacing). Each row is one gate, amendment or attempt. Grey squares, process steps (adopted, opened); red cross, failed; amber diamond, not evaluable, not run, infeasible or futile; green circle, GO in simulation only (not a gate). A1 (tail-G2, G1, G3 on frozen v1): failed; a post hoc contamination explanation is unconfirmed. A2 fresh nulls: not evaluable (6 of ≥10 projects admitted, after a re-walk to fix an implementation bug). IT-CID nulls: futile. A3 (v2_sep tiers): expected not evaluable. A4: not run (simulated P(v1 minimum) 0.08–0.43). C9 pilot: GO by simulation. A5: knowingly relaxed the claim conditions to allow a single T1 + T5 look. A6/HN1: not tagged, infeasible (probability 0.63 < 0.8 at the measured throughput). Sealed look (opened 28 October 2026, 14:23 CET): T5 completed; T1 was killed when the compute node was wiped, and the look was recorded as failed (unterminated) with outputs deleted unread. Up to two retries from the frozen commit remain allowed under A5. No null-spectrum gate was passed, and no held-out number has been seen.

Supplementary information (planned)

- **Supplementary Table 1.** Datasets: identifier, accession, role (training, development, null, sealed), instrument, number of runs, licence and use in this study.
- **Supplementary Table 2.** Pre-registration and amendments A1–A6: date, tag or commit, content, and outcome in the decision log's words.
- **Supplementary Table 3.** Per-workflow accuracy and runtime on development data, including cross-linking (D3, D3b), LiP (D7), co-fractionation (D10cf), open and mass-offset search, MBR false-transfer rates (D9, D12) and TMT.
- **Supplementary Table 4.** Full gate table (G0–G5, tail-G2, arm M, entrapment verdicts per dataset and level).
- **Supplementary Table 5.** Negative results and abandoned approaches, with the experiment that tested each.
- **Supplementary Table 6.** Timing records with hardware fingerprint, load average and lock status for every speed number.